LINUX SERVER SENSOR TRANSCRIPT

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L00 - INTRODUCTION

L00A-PRES: Introduction (03m:05s)

00:00-01:00

Hi and welcome to the *Stellar Cyber Open XDR Linux Server Sensor* module. I'm David Shirley from Stellar Cyber Enablement and Training teams. The lessons in this module show you how to install, configure, and verify a Stellar Cyber Linux Server sensor deployment. We will cover the software installation on a Linux Server and authorization of the server sensor in the Stellar Cyber Data Processor. We will cover validation of the installation and

configuration, as well as advanced topics including File Integrity Monitoring, and network traffic features for the Linux Server sensor. Throughout this module there may be new concepts and terms that you may not be familiar with, but that's okay and to be expected. The important thing is that you become familiar with the concepts and terms described. Through the Stellar Cyber Open XDR training curriculum, your understanding and application of cybersecurity principles will deepen with each repetition, iteration, and example

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as you advance on your way to becoming a Stellar Cyber cybersecurity Superstar. Thank you for your dedication.

Welcome to Stellar Cyber Open XDR Linux Server sensor installation and configuration. The following lessons are included in this module:

- *Overview*. At a high level we will see where the Linux Server sensor fits into the Stellar Cyber platform, some of the prerequisites based on different network architectures, and confirm resources required.
- *Software Installation*. This lesson takes you from acquiring the Stellar Cyber Linux Server software to installing, configuring, and validating the installation on the Linux Server host.
- *Server Sensor Authorization*. With the server sensor software installed on the host,

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it's over to the Data Processor's user interface to continue by authorizing the sensor, but first we will create a custom sensor profile to use.

- *Server Sensor Validation*. This lesson validates the installation and configuration using both the Windows host and the Data Processor's UI. By the end of this lesson, we will see Interflow records from the logs on the Linux Server in the Data Processor.
- *Enhanced FIM*. File Integrity Monitoring is an important element of maintaining the security of Linux Servers. This lesson focuses on enabling the feature and validating the configuration through testing.
- *Enhanced Network Traffic*. The Linux Server sensor includes a network traffic monitoring feature similar to that of a Modular Sensor. This lesson will show the differences and also the configuration and validation of the network traffic feature.

Now let's get going and happy learning!

L01 - OVERVIEW

L01A-PRES: Overview (10m:05s)

00:00-01:00

The first lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Overview. Overview.* In this lesson we take a high-level look at where the Linux Server sensor fits into the Stellar Cyber platform, discuss some of the prerequisite based on different network architectures, and confirm resources required.

First let's start with the big picture. In the top right-hand side of this diagram we have our Stellar Cyber Open XDR platform. That platform is made up of a Data Processor including a Data Analyzer and a Data Lake. It's where the Interflow records are stored, normalization occurs, enrichment. We have different types of detections whether it's supervised machine learning or unsupervised machine learning. We have heuristic analysis rules, ATH playbook rules,

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graph machine learning correlation. We can do search, of course the user interface, and our own API to pull information in and out of that. What we're most interested in is our Data Processor made up of our Data Lake and our Data Analyzer. On the left-hand side is our Modular Sensor. The primary role for the Modular Sensor is to collect data from network and security devices on that network and forward to our Data Processor. The Modular Sensor has features including log forwarding, Network Traffic, IDS, and Malware Sandbox. It can have a feature as an Aggregator, which we'll talk about during this module. Tenable Nessus scanner and connectors can also be run through our Modular Sensor. Buffering is available in some of these features, for when communication to the Data Processor is interrupted. We can buffer some of that information, hold it, wait for the connection to come back, and then send it through.

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In the middle of our diagram of our network we have our server sensors. We have Windows Server sensors and Linux Server sensors and this lesson and module is focused on the Linux Server sensors in our diagram. We're going to look at how to install it, how to configure it, and how to verify it. We'll look at two of the advanced features. We'll be looking at File Integrity Monitoring and we'll also look at network traffic from the Linux Server sensor's perspective. The Linux Server sensor is Stellar Cyber software installed on physical or virtual servers, either on prem or in the cloud, to collect data that is used to detect threats on Linux Servers. What we're going to be paying attention to is the software that's installed on that

server, how it communicates with the device to pull the logs and events, and also that north-south traffic.

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We will also talk about that. We will talk about how to configure it, some network architecture layouts, firewall requirements, so let's continue on. The Linux Server sensor gathers data from within the Linux Servers and generates Interflow records from this data, which is then forwarded to the Stellar Cyber platform. Linux Server sensors monitor and collect the information, process info, command execution, files, file events, and network traffic metadata. The server sensor converts that information to metadata and forwards it to the Data Processor as an Interflow record. The Data Processor can then correlate the traffic processes, users, and commands for security, dynamic denial of , and breach attempt detections.

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Here are some prerequisites for our Linux Server sensor. Xeon two virtual cores, 12 gig of RAM, and 128 of SSD disk. We will look at the firewall ports required to open to communicate from this Linux Server sensor to our Data Processor made up of a Data Lake and a Data Analyzer, and then we'll also look at some Python requirements that are different depending on your target environment. Here are some of the processes that you will see running on your server sensor as a result of this installation. We have our Aella audit (aella_audit) and that collects the audit logs and provides the File Integrity Monitoring we'll see that when we turn that on. Aella Conf (aella_conf) handles the configuration. When we push down the configuration for our server sensor coming from a sensor profile, that is the one that handles the configuration. Aella control (aella_ctrl) monitors other s, and can start and stop them based on the configuration.

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Aella Flow (aella_flow) collects the metadata in traffic, so that's doing our network traffic that's collecting. Aella Mon (aella_mon) collects system resource usage, including the CPU, RAM, and disk. These are all launched following the installation of our Linux Server sensor software. There are a couple of different network architectures. This is the more simplified one. We'll start with this where we have on the left-hand side our Linux Server sensors, a firewall in the middle. Actually the most simplified one would be no firewall at all, in which case we don't have to worry about firewall rules. But in our example here we're going to talk about having a firewall in the middle and what's required, what ports need to be opened on that firewall for our Linux Server sensor to communicate with our Data Processor, which is made up of that Data Lake and the Data Analyzer. Let's have a look at those in a table format.

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In the left-hand column we've got our Linux Server sensor, the second column, where it's communicating to, so Data Lake is the DL and Data Analyzer is the DA. Starting in the top row our server sensor needs to communicate to the Data Lake. We need to have ports 6640 through to 6648 open, and they need to be open with a protocol of TCP for TLS 1.2. That's for communicating with the CM, the configuration manager. The second row: Linux Server sensor again talking to the Data Lake on 8443. Again, protocol TCP with HTTPS with TLS with TLS 1.2. That's for downloading software and files from the DP including custom parcel logs. If we go to upgrade this server sensor, that is the requirement in your firewall to do that. The final one is the Linux Server sensor communicating to the DA, the Data Analyzer, to put the two ports that need to be open there is 8888 and 8889 again, running protocol TCP HTTPS with TLS 1.2.

07:00-08:00

That is for the receiver port communicating with the Data Analyzer. In that first network diagram, these are the ports that are required to be open. In the second network architecture, we could probably actually have three here. Right? We could have what we've added here is a second firewall. We've got our Linux Server sensor on the left, a firewall, a Modular Sensor in the middle, and another firewall before getting to our Data Processor. The difference here is that we're going to configure that Modular Sensor in the center with the Aggregator feature turned on. What that enables us to do is to have all our server sensors, whether they're Linux or Windows, other Modular Sensors on that left-hand side communicate through the firewall to the Modular Sensor. Then on the firewall on the right the only ports that are required to be open for the Modular Sensor to communicate with the Data Processor

08:00-09:00

are ones specifically for that Modular Sensor. The firewall on the right doesn't need all of the rules open for all of our different sensors on the left-hand side–Linux, Windows, Modular Sensors–if they are communicating via that Aggregator. Now in our environment that we'll see in the demo system, the Modular Sensor and the Linux sensor happen to be on the same network. That firewall on the left doesn't exist. But for the purposes of explaining this and showing you these tables in the next couple of slides, let's have server sensor on the left firewall, Modular Sensor with the Aggregator feature turned on in the middle, firewall, and then on the right-hand side our Data Processor consisting of our Data Lake and a Data Analyzer. This is for the Linux Server sensor to communicate only with the Modular Sensor with the Aggregator feature turned on.

09:00-10:05

These are the firewall ports required for that left-hand firewall. You can see here they're all bundled up together. 6640 through 6648 and then 8443 and 8888 and 8889. They were all the ones we saw before. For TCP proxy, those would need to all be open on that left-hand firewall for communication between the Linux Server sensor and our Modular Sensor with the Aggregator feature turned on. Other ones required, again, depending on your network environment, for our DNS environment and NTP. NTP is crucially important in this security space for making sure our time is synced. Depending on where your NTP server is or what your DNS looks like, again, our Linux Server sensor needs to be able to communicate via port 53 to get to DNS, and via port 123 to get to an NTP server.

The *Overview* lesson is complete. Thank you.

L02 - SOFTWARE INSTALLATION

L02A-PRES: Software Installation (04m:44s)

00:00-01:00

The next lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Software Installation. Software Installation.* This lesson covers acquiring, installing, and configuring the Stellar Cyber Linux Server software on the Linux Server host.

The very first thing we need to do is identify the version of Linux that you're running. The best thing to do here is to go to our Knowledge Base at docs.stellarcyber.ai. You can see there on the left-hand side we're going to browse to **Installing** and then **Installing a Server Sensor**, and finally **Installing a Linux Server Sensor**. Scroll down to where you'll find the different types of Linux supported and what the installation for that version of Linux would look like. You can see in that top menu bar we are looking at the Knowledge Base for 4.3.7.

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You can click on that drop down and choose the version you are running in your deployment, then go find the version that you need for Linux. For this example, we are running 4.3.7 is the Stellar Cyber version that we want to install, and our version of Linux is Ubuntu 20.04.6. That is the version that we're running, so let's continue. We will have a curl command, and the curl command requirement is all spelled out there. That comes from the documentation. What you're required to change in that command is the login and the password. You can see at the very beginning, you'll need to contact Stellar Cyber's Customer Support at support@stellarcyber.ai to get those login credentials. The other thing you have to substitute further down the path you'll see a /release/version. You need to change that word, “version” to be 4.3.7.

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We'll see an example of this. Here is our curl command on our Linux host. You can see our curl -k -u. We've blanked out our username and password that you'll need to get from Stellar Cyber support. You can see at the very end of that we've got /release/4.3.7. All that's going to do is pull down the script required for our version of Linux to then run the script to pull down the software. Here it is running. Here's our script running. We've got a `sudo bash ds_Linux_install.sh-v` and then the version. Again, we must substitute the word, "version" for the version of code we're running. In our example, 4.3.7. You can see it pulls down the software and starts installing it. What we're looking for at the end of it, the confirmation is that the Stellar OS modules are all running.

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Once that is all finished we can issue the command `aella_cli`, and that will run our server sensor software and give us a command prompt for the command-line interface of our software. Run that command, and then the first thing we need to do is set our CM. No different to our Modular Sensor. We set a CM to the IP address of the management interface of our Data Lake. In this example, it's a `set cm` to 192.168.1.175 The next thing to do, once you've set that, is to check that it has worked and you've got communication verified. Do a `show cm` and what we're looking for is one that we've typed in the right address our 192.168.1.175, and the second piece is we're looking for that status of ESTABLISHED.

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You can actually see if you run that `show cm` command really quickly after issuing the `set cm` command, it may not have had a chance to communicate with our Data Processor and had a chance to become established. If it doesn't work at first, just wait a couple of moments, run the `show cm` command again until we see the established is what we're looking for. Not too much to installing, downloading, and installing our server sensor software, and configuring it to talk to our Data Processor. The next step is to go into our Data Processor and verify and authorize.

The lesson, *Software Installation*, is complete. Thank you.

L02B-DEMO: Software Installation (06m:35s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Software Installation Demonstration. Demonstration of Software Installation.*

In this demonstration, I will show you the installation, configuration, and verification of the Stellar Cyber Linux Server sensor, including downloading the installation script, installing the software, and configuring the Linux Server sensor for communications with the Stellar Cyber DP. Let's start with the Linux Server details. Okay. Here we are. Let's log in. Alright. Here we are in our Linux host, our Ubuntu host.

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Let's see what version we are running. Alright. What we can see here that we really want to look at is down at the bottom. There is Ubuntu 20.04.6. Twenty zero four is the version we're running of Ubuntu and that would be important when it comes to how we run the scripts. Let's check out our host IP address. It's 10.20.1.23. Okay. So far so good. Let's go to our documentation and look at the version of Ubuntu 20.04 that we've got. So a couple of different ways we can either go up here into the top right and search the Knowledge Base for installing a Linux Server sensor or I can go over here, expand **INSTALLING**, expand **Installing a Server Sensor**, and there it is the first one: **Installing a Linux Server Sensor**. I select that one.

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Let's scroll down and find our Ubuntu 20.04. There we are. It'll bring us down here. The first thing we have to do is run this curl command. You need to substitute the login name and the password, those two credentials, which you can get from Stellar Cyber support. Send an email to them at support@stellarcyber.ai for those credentials. The other piece of information you need to change in this curl line is the version number. In our example, we're going to use 4.3.7 so that we can show you in a later lesson how to upgrade the server sensor software from 4.3.7 to 5.1.1. Let's type this command in with the login and password and version substituted. Okay. Back to our host.

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Run our command. There is our script downloaded. Fantastic. The next thing to do if we go back to our Knowledge Base, we can see the next thing to do is to run a sudo command. Again, we need to substitute out the version at the end of that. We will substitute that with 4.3.7. I'm going to copy that in. Let's watch that download after a password. Let's wait for that server software to be pulled down. Here it comes. Fantastic! It's all down. It's been installed. We can see that they are running, installing, and shutting down s. Just about there.

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The last thing we want to see is that the s are all up and running. There we go. We've got our s up and running. We have our command prompt back, so we should be able to type in `aella_cli`. We are in our software, our server software. The first thing we need to do is

configure it to talk to the Data Lake. To get to our Data Lake, we need to know an IP address. Let's just confirm that in the UI. Here we are in the UI of our Data Processor. **System**, under the **Data Processor** grouping, we go to **Data Lake**. In our table, if we expand that out, we can see our name of the cluster is sc-dl-cluster and the IP address, which is 192.168.1.175. Back to our console. Let's just do a `show interface` to see.

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Just so you know, what I can do is type s-h-o and press tab. If you type in at least three characters of the command, you can then press tab and it will fill it out. `show int` press tab and it will spell it out for you. We can see our interface: 10.20.1.23. That's the IP address of the Linux host we're configuring. We're all good there. We just need to `set cm` of that management IP address of the Data Lake. It's `192.168.1.175`. Alright. That's been accepted. If we do this really quickly, if we do a `show cm`, we'll get an error. Because we've made a change to the configuration, that change has now been stored and services are restarting. We have to be just a little bit patient while we wait for those s to come up again, and we can just run that command again to see the output that we're expecting.

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Just wait a few moments till we'll run the command again. Alright. There we go. We `show cm` of our cm controller, our 192.168.1.175. The great news is the **Status** is ESTABLISHED. We're ready to do the next piece of our configuration, which is the authorization from the UI. We will do that in our next lesson.

The lesson, *Demonstration of Software Installation*, is complete. Thank you.

L03 - SENSOR AUTHORIZATION

L03A-PRES: Sensor Authorization (04m:55s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Server Sensor Authorization. Server Sensor Authorization.* In this lesson, we will create a custom sensor profile.

Now that we've installed our Linux Server sensor software on our Linux host, we have configured the CM to point to our Data Processor, we're ready to authorize it in our Data Processor. Here we are in the UI. We go to **System | Sensors** to see if it has been added to the sensor table. Here we can see it. We can see that the hostname of the Linux host is what

the server sensor name is in our table. We can also see that from an authorization perspective, it is not authorized. We've only just put it there.

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We're happy that the status is green, the status is the green dot, but from an authorization perspective, it's not authorized yet to send any data to this Data Processor. You can see there when we add a sensor for the first time to the table, it picks up a default sensor profile. In this example, because it's a standard sensor profile for a Linux Server sensor or a Windows Server sensor, it picks up the sensor profile of default. If it was a Modular Sensor, it would pick up default modular. Just notice it's the sensor profile that picks up and we don't want to use that. We always want to be proactive about the sensor profile we use depending on your deployment. Maybe you want to use a different profile for your Windows host, a different profile for your Linux hosts, a different profile for different types of servers. Our recommendation is that you create a server sensor profile.

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Let's go and do that. Here's how we do it. We go to **System** and then **Sensor Profiles**. We click the **Create** button over to the right-hand side. From the drop down, we're going to add a standard sensor profile. For our server sensors, Linux and Windows, they use standard sensor profiles. If it was a Modular Sensor, it would use a Modular Sensor profile, but it's a Linux host. Standard profile it is. The next pop-up window is how you add and configure it. Give it a name. That is a name of your choosing that makes sense to you in your deployment. We're going to then choose the default receiver. We'll talk about receivers in another lesson. It's a Linux host, so we're going to click the **Agent** tab in this example, and we're going to expand the **General** section. When you expand the **General** section, it will expose the Linux auditing.

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Enable that slider in for Linux auditing and click **Submit**. Right. We've created our new sensor profile. We need to go and bind our sensor to that profile. We navigate back to **System** and **Sensors**. We find our sensor listed and click on the pencil to edit that. When we open it, we can see that there is a drop down for sensor profiles. It's currently at Default. We want to hit that drop down and choose our newly created sensor profile. After we've done that, choose **Submit**. Okay. We can see in our list now. We've got our sensor. Its **Status** is green, which is the equivalent of established. But we can still see it's not authorized. We can see in that **Sensor Profile** column on the right-hand side that it is using the Training Server Sensor Profile.

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The next thing to do is to make sure we authorize this sensor to be used in our Data Processor. From the top right, we click **Manage**, from the drop-down **Authorization**, and we will get a new pop-up window. Our unauthorized sensors are now going to be on the left-hand side. You can see there in the top of this window that radio button is on **Authorize**, so we're getting sensors that are currently unauthorized to make them authorized. Check the box next to our sensor that's unauthorized. The next thing to do is hit the **Add** button to add it to the table on the right. We can see it's now in the table on the right. All that's left to do is click **Submit** to authorize that Linux Server sensor.

The lesson, *Server Sensor Authorization*, is complete. Thank you.

L03B-DEMO: Sensor Authorization (06m:40s)

00:00-01:00

The next lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Demonstration of Server Sensor Authorization. Demonstration of Server Sensor Authorization.* With the server sensor software installed on the host, it's over to the Data Processor's user interface to continue authorizing the sensor.

In this demonstration, I will show you how to complete the Linux Server sensor installation by authorizing the sensor and changing the configuration of the sensor to use a custom sensor profile rather than the default sensor profile. Let's start with the Linux Server sensor CM configuration. Here I am back in our Linux Server. The last thing we did was a `set cm` and a `show cm` and a `show interface`.

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Let's make sure that's still the case. `show cm 192.158.1.175` and it is ESTABLISHED. The next thing to do is to go into our user interface of the Data Processor and authorize this sensor. But first, we're going to change the profile that the sensor is going to use. If we have a look at the profiles that we've currently got, let's do this. Let's go **System | Sensor**. First of all, is our sensor here? Yes, it is. Fantastic. If we make that a little bigger, we can see that our Linux Server sensor is there, our ublnx-sj-tr-dbw3. That's our server. We know it's 10.20.1.23. It's got a green light. That's the equivalent of established in our CLI, but in our UI we see a green dot. What we are seeing here is the “Not Authorized.”

02:00-03:00

We're required to authorize it before any data will be sent backwards and forwards. But what we can see here because it's just been added to this table. It's added using the Default sensor profile. If we make that column a little bigger, our Sensor Profile. By default, when it gets added, it uses the default profile. If we're to go look at the profiles available, sensor profiles, we can see that there is a default sensor profile, which is a standard sensor profile

type. That is used for both Windows Server sensors and Linux Server sensors. Then we have a default_modular. Its profile type is a Modular Sensor. Of course it's used for Modular Sensors, both virtualized and our physical photons. Then we can see we've got another standard sensor profile that we used previously with our Windows server sensor. If we were to create a new one, it's very simple. It's just create.

03:00-04:00

We're going to create a standard sensor profile because that is what our Linux Server sensor uses. We give it a name. This could be a test name of David, for example. We choose a default receiver, and that's it. We're ready to go. All we'd have to do then is scroll down and submit. I've now created a new–what did we call it? We called it David. There is our new standard type of sensor profile. We don't need to use that. We're going to use the one that already exists. I'm going to hit the trash can and okay. That's gone. We're going to change our Linux Server sensor to use this Training Server Sensor Profile so it can share the same profile. Now maybe you want to create one profile for your Windows Server sensors and a different profile for your Linux Server sensors being that they don't overlap inside of the profile. Actually let's just go in there and show you that. Add standard sensor profile.

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We can see here we've got (a tab called) **Agent**. **Agent** is for our Linux. **Windows** is for our Windows. They don't really overlap. It'd be fine in some circumstances to use this. Maybe you're going to have different Linux servers to Windows servers. You want to split them up just to really know what you're going to use, but we're just going to use the same one here. Back to our Linux Server sensor, so **System**, and **Sensors**. Here is our server here. We're going to edit that server by clicking on the pencil. We're going to change the sensor profile, click on the drop down, and we're going to choose our profile. You can see the only profiles that show up, of course, are the standard profiles. We wouldn't see Modular Sensors here. We're going to select our **Training Sensor Profile**, and then select **Submit**. That will make that change. Now we can see here is our Linux Server that's not yet authorized that is now using the Training Server Sensor Profile.

05:00-06:00

The next thing to do is **Manage | Authorization**. We can see on our left-hand panel here, it's the only server ready to be authorized so we select it. Add it, and now it's in the list of authorized server sensors, and we click **Submit**. You can see it's gone red. It's disconnected. What's happening is the configuration that's in this sensor profile, our training server sensor profile. All of the configuration that's in this profile is now being pushed down to our Linux Server sensor. We need to wait a few moments for that to happen. While we are waiting you can click refresh, on the table to see what's happening. We could go back to our

console and `show cm`. Let's see if that's happy. That is still happy, which is great. Let's come back to our UI and then refresh.

06:00-06:40
Now we've got a green server sensor. The status is green. We are now authorized. We are ready. You can see over here where we're talking about earlier. The reason we chose 4.3.7 is we have an opportunity now. It shows you that there is an upgrade available for our 4.3.7 Linux Server sensor. We'll do that at a later time. There we go, up and running, ready to ingest some data.

The lesson, *Demonstration of Server Sensor Authorization,* is complete. Thank you.

L04 - SENSOR VALIDATION

L04A-PRES: Sensor Validation (06m:30s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Server Sensor Installation and Configuration* module is *Server Sensor Validation. Server Sensor Validation*. In this lesson, we validate the installation and configuration using the Windows host and the Data Processor's UI.

To recap, we have gone to our documentation to see which version of Linux we're running. Ubuntu 20.04 was our version. We used that to decide on which curl command to pull down the installation script. We ran the installation script, changing the username and password and the version number of what version of code we're downloading and what version of software. We've run the `installation` curl command to install the software, we've run our `aella_cli` command to configure our server sensor with a `set cm` command.

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We validated that. Then to the UI to validate that we can see the connection of our server sensor to our Data Processor. We created a new sensor profile that we have now added to our server sensor, and then we have authorized that. We've done all that. Now we're going to validate that all of that ends up and running. Let's first do that. We're going to do it in two ways. We're first going to look at our server sensor, on our Linux host and then we're going to go into our UI and validate it in there. The first thing to do in here is run `aella_cli` and let's clarify a couple of things. The first one: a `show cm`. We just want to make sure that one, it's correct. I think it's pretty hard for it not to be correct here if we've authorized it in our UI, but we can see our controller here is 192.168.1.175.

02:00-03:00

What we're really looking for is the status is ESTABLISHED. The next thing to do, we haven't run this before `show receiver`, but this is interesting when it comes back to thinking about the firewall settings we had to have open. The `show receiver` command shows the receiver for the DA of 192.168.1.150. We didn't set that command. All we did was a `set cm` of that 192.168.1.175 because of our sensor profile that uses the default receiver and the receiver is configured for our Data Analyzer of 192.168.1.150 when that configuration is pushed down to our server sensor. We can now see that we've got that established for a `show receiver`. We want to see status enabled there. Then the last piece we want to see is that a `show system` will show us that all of our s are running. That's what we're expecting to see. That's the configuration validation on our Linux host.

03:00-04:00

Let's go to our UI. Inside our UI, we'll start with a **System | Sensors**. It'll show us the table. What we're expecting to see in this table, one is the green dot. That is the equivalent in the last slide of Status: ESTABLISHED in the `show cm`. We want to see that green dot. The next one we want to see is authorized because we have authorized this server sensor going through the manage authorization process. Then the next thing I'm going to do is actually click the server sensor's name to see a bit more information about this. Number one: We're going to look over in that right-hand column in the server and status and we want to see a bunch of green dots. So far so good. What we're seeing there. The next one is what is the profile bound to it? We can see that we've got our profile that we created that we expect to see bound to our sensor.

04:00-05:00

The next one is what are we looking at from a Linux agent output? This is the collection of the logs from the Linux host. We can see a bunch of those coming in there. The next piece to see here is our Interflow status. The **Interflow Stats**, sorry. The **Interflow Stats** is both input and output. You can see that input is a much bigger number–that twenty five point eight million versus two hundred and nineteen point four thousand. The difference between that is that we're not actually sending all the data that we're seeing from that north-south traffic back to our Data Processor. We're sending metadata of that. Again, the metadata converting that to an Interflow, and that's what's been sent. That's the output is why it's so much smaller. That last piece, the reduction ratio. If you did the calculation of twenty five point eight million divided by two hundred and nineteen point four thousand I think you'll find it is a hundred and seventeen times.

05:00-06:00

That's the reduction rate in what we're seeing there. There's our clarification from our sensor's perspective. Let's look a little further and see actual data coming in. Let's start there with an **Investigate | Threat Hunting**. We know we need to change the indices from

Alerts to **Linux Events**. It defaults to **Alerts** every time we go to **Investigate | Threat Hunting**. The default index here or indices is **Alerts**, so we need to change that from **Alerts** to **Linux Events**. There is our output. We can see our sensor listed in that top left-hand corner. That name is the hostname of our Linux host but it's actually the server sensor name is what we're seeing there. We confirm that. Well that's there and there are 244 records so that Linux events coming in from that perspective. We can look at the documents over time, which is a great one.

06:00-06:30
You can see from a time perspective, the volume of events that are coming in. We've only just turned this on, so we don't expect to see that fully populated. Then down the bottom we've got our table. We were actually seeing the individual Interflow records with those Linux events. We can see from this page that our configuration has been validated.

The lesson, *Server Sensor Validation*, is complete. Thank you.

L04B-PRES: Sensor Validation (10m:00s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Demonstration of Server Sensor Validation. Demonstration of Server Sensor Validation.* This lesson demonstrates the validation of the installation and configuration using both the Windows host and the Data Processor's UI. By the end of this demonstration, we will see Interflow records from the logs on the Linux Server in the Data Processor.

In this demonstration, I will show you how to validate that the Linux Server sensor software installation and configuration was successful by looking at ingested data on the Data Processor from the Linux Server sensor. Let's start with the Linux Server sensor software. Here we are. We have logged into our Linux box.

01:00-02:00
Just to show you in and out of this. Entering `quit` will take us back to the Linux prompt. Enter `aella_cli` to get us back into our prompt. If we do a `show cm` we see ESTABLISHED. Fantastic! 192.168.1.175 that is our Data Processor. That is all good. We haven't run this before. Let's do a `show receiver` command. Now we can see that receiver is pointing to 192.168.1.150. Now we didn't configure that, so we'll go and show you where that shows up in the Data Processor. Into our Data Processor, if we go to **System** and under **Data Processor**, we go to **Data Analyzer**. We have a Data Analyzer called sc-dp-da-master, and there is its address: 192.168.1.150.

02:00-03:00
That was all part of the firewall rules that we had to make sure we had open that there is communication from our Linux Server sensor to both the DL, the Data Lake, that 192.168.1.175, and our DA, our Data Analyzer, here 192.168.1.150, which we saw when we ran that command `show receiver`. Okay. So far all good. Let's go back to our sensor in the UI and have a look. **System | Sensors**. We're still green, and we are authorized, all good. If we click on the name of our sensor we can see some details in here. Details we can see the name in there, our sensor ID, what software version it's running. It's a Linux Server sensor. What it is running, the OS is Ubuntu 20.04. We saw that Our IP address confirmation 10.20.1.23 running in VMware. We're using the Training Server Sensor Profile as a sensor profile.

03:00-04:00
Up in the right-hand corner here under **SERVICE** we can see all of our services are up and green. If we come down here and look there's probably a little something not right here–a **Linux Agent Output** of zero. That's not what we're expecting. We're expecting to see log information coming from our server. Let's jump into **Investigate | Threat Hunting** and have a look in there. Now we know it defaults to **Alerts**, so let's change it from **Alerts** to **Linux Events**. Here we are at **Linux Events**, and we've still got a big fat zero. That's not encouraging. We know our Linux Server sensor software also runs as a traffic monitor. Let's change from the **Linux Events** to **Traffic**. Okay. We've got a bunch of traffic coming in. If I'm to change our **Traffic** to say I only only want to see the **Linux Agents**

04:00-05:00
so I'm going to filter for Linux Agents, and I see our server here. Our server is very happily sending data to our **Traffic** index, but we know when we go look, let's look at all of our indexes. We're only looking at the last one hour, so it's fine. We could **Select All** indexes. Wow. We are getting half of what we expect within data coming to our **Traffic** index, sensor monitoring index. Of course, it's a sensor like a Modular Sensor. It's a Windows Server sensor, so there's data coming into the **Sensor Monitoring** index. It otherwise all looks happy and okay. Let's go back and check one more thing on our server sensor CLI software. Back to here and let's run a `show system` and see what that results for us.

05:00-06:00
Well, there we go. What we're expecting to see is an aella_audit, which is what would be bringing in our Linux log and that is stopped. Now how do we fix that? Let's go have a look back in our UI. If we go to **System | Sensors**, we know this is our sensor here. Expand that out. That's our sensor. If we go across, this is the sensor profile it's using. If I click that, I can edit it. I come into **Agent**, which is for our Linux. Well, we haven't turned on **FIM Linux** yet.

We'll do that in a later lesson. That is our File Integrity Monitoring for Linux. But if I click the drop down to expand **General**, **Linux Auditing** is not turned on. I hit that slider to turn it on and submit.

06:00-07:00

Immediately, both our AD3SRV, our Windows server, and our Ubuntu server go red because they're both using the same sensor profile. They're both using that Training Server Sensor Profile. It won't take long. Let's do a refresh. They're back up in green. If I go back to the console, let's just do a `show system` again and look for aella_audit. We want to see that that's running. There we go. It is running. Back to the UI. It's only been a few moments, but let's see if there is any traffic coming in. Let's go back to the beginning of looking at our Sensor. It's still zero. Maybe we just need to wait a little bit. What we can see here is that that is now running. We now have **Linux Events Logs** with a green status. That was not there before. It's just a matter of time for those events to come in. Have we got a filter?

07:00-08:00

We do have a filter, but that shouldn't affect us in here. We'll just have to wait a few moments for this to start seeing some logs from our Linux agent. Let's just wait a moment or two and I'll come back and have a look. In the meantime, we could go to **Investigate | Threat Hunting**, change from **Alerts** to **Linux Events**. It's still zero. We're just waiting for those to come through. I'm going to go do a couple of things on the Linux box, maybe log into it and log out, just so we see something come through, so standby. Alright. What I just did is I just went and logged in and logged out of this device. If I can do a right-click and I come back in here. I log in as `scengineer`. Here I am logging into that Linux host and let's log out straight away.

08:00-09:00

I come back to my UI, **System | Sensors**, open up our sensor and we can see already I only got a whole bunch of output. It was just built up waiting to come through. There we go! If I now go and this is just in the last hour Let's go back to **Investigate | Threat Hunting**, change from the **Alerts** index to the **Linux Events** index. Here we go! I have got a bunch already coming in. Let's go have a look. You can see where it came from. Let's go look at some of the fields that are interesting to us. We've got the computer name over here. Actually, it might be a little easier to click on the "**i**" and look at **Details**, and we can scroll through this way and see what we can see. So our host name, one thing we can see, the agent type is auditbeat, our computer name, our host IP address.

09:00-10:00

We can see under all of these **host_geo** that's all our enriched fields. The message class is an audit rule message origin source, Linux agent. We can see that our agent is in the root

tenant. There we go. All we have to do is reconfigure. Navigate to **System | Sensor Profiles**, go into our profile, edit it, go to **Agent** and then make sure in general that **Linux Auditing** was turned on. There we go. We have configured our software. We have validated our software and now we can see data coming into the system.

The lesson, *Demonstration of Server Sensor Validation,* is complete. Thank you.

L05 - ENHANCED FIM

L05A-PRES: Enhanced FIM (06m:50s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Enhanced FIM. Enhanced FIM.* File Integrity Monitoring is an important element of maintaining the security of Linux servers. This lesson focuses on enabling the feature and validating the configuration through testing.

Stellar Cyber includes a File Integrity Monitoring feature for both Windows Server sensors and Linux Server sensors that lets you keep track of changes to specified files and directories on the host server sensor machine including file changes, file creations, and file deletions. Configurable fields include: **Enabled** - the ability to leave the entry in the table but choose to enable or disable that entry.

01:00-02:00
The **Path** - the directory that's being monitored. **Inclusions** is what files are being monitored. You can see you can use wildcards there. **Exclusions** - what files are not to be included in the monitoring in that same path, and **Depth** - how many sub directories to monitor below what's the directory that's entered into the path. For our example, we'll make a directory called FileMonitorTest and we're going to add that path as the directory that we're going to monitor. You can see the `mkdir` to create the directory, and allows us to prove that the directory is there. The PWD, so the present working directory of /home/scengineer/File MonitorTest. It is case sensitive, so those capital F, capital M, and capital T are important.

02:00-03:00
Back to the UI. FIM is configured in the sensor profile. It's not specific to the sensor. It's the sensor profile to which the sensor is bound. To enable FIM, do the following steps. Go to **System | Sensor Profiles** and click the pencil to edit that profile. After you're in the profile, because we're editing FIM for our Linux Server sensor, we will use the **Agent** tab. We can see when we select the **Agent** tab, there is FIM. It is off by default. We just click on the slider to turn it on. You can see that there are already two pages of default folders to be monitored.

Those were created by Stellar Cyber and you can choose to leave them there. You could delete them. You could uncheck them so they're not enabled and part of the monitoring service. That is all up to you.

03:00-04:00

We want to add a row. We're going to add a row for our /home/scengineer/FileMonitorTest. Select **+ Add Row** to add the row. That'll create a new row. Then in the **Path** column we go and paste in. If we've copied it or type it, I always figure it's better to paste it to make sure you've got it exactly right. We'll leave the defaults of include everything, exclude nothing, and a depth of ten and we're going to monitor for changes, for creations, and for deletions. After you've done that, select **Submit**. We know that once we press **Submit**, we've made a change to the sensor profile. Any server sensor that is bound to that profile will have this new configuration pushed down to it. It'll momentarily change the configuration for that server sensor. Back to test it. After we get in there, let's test. Let's create a file so we can see we're in the correct directory with a PWD, a present working directory.

04:00-05:00

We see there that we're going to create a file called david.txt, we do a `ls -la` to prove that's files there, and we've just added some sample text in there. So `nano david.txt` to create the file type, add in some sample text, save the file, and that is complete. We've made a change to a file that is in a directory that's being monitored by the FIM feature. In the UI we can go to **Visualize** and there is a dashboard that's already created called **File Integrity Monitoring**. Inside of that we've got a couple of charts that we can see. One is a counter at the top is how many additions there's been to the File Integrity Monitoring. You can see three. There's a bar chart on the right-hand side, which is our documents over time to show us what it looks like from a time perspective. In the bottom left it will show us a split in that pie chart between changes, creations, and deletions.

05:00-06:00

That's what we can see there. Observe that we've got a couple of create and change actions there. The other place to monitor this is in **Investigate | Threat Hunting**. We know if we go to **Investigate | Threat Hunting** we need to change from the **Alerts** index to the **Linux Events** index. After we've done that, we're going to enter a search. Now this is a pretty specific search and it's using some regular expression. It's a little odd in what it looks like because you would say the file path was /home/scengineer/FileMonitorTest. But when we put it there as a search, we have to be careful of what those forward slashes mean. A backslash before the forward slash actually is literal, so be literal with the next character. We do file_path: and in quotes we have backslash, meaning include the forward slash literally as a forward slash character

06:00-06:50
because that's how the path is known. Then /home. The same thing the backslash before the /scengineer is the same thing. These are regular expression way of searching. Ultimately, what we're looking for is a file path with home sc engineering file test monitoring. After we've typed that into our search window we can see in the table below, we get both. We've put two columns in here: a column saying which is the rule in our FIM configuration, in our sensor profile and what was the result that hit that. That's the other place. We can verify that our configuration for FIM is working.

The lesson, *Enhanced FIM*, is complete. Thank you.

L05B-DEMO: Enhanced FIM (07m:25s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Demonstration of Enhanced FIM. Demonstration of Enhanced FIM.* File Integrity Monitoring is an important element of maintaining the security of Linux Servers. This lesson demonstrates enabling the feature and validating the configuration through testing.

In this demonstration, I will show you how to enable the FIM feature or the File Integrity Monitoring feature for Linux Server sensor. Let's start with setting up a test on the Linux host. We're logged in. First, let's make a directory: (types) `mkdir FileMonitorTest`. There is our file monitor test.

01:00-02:00
Let's go into that directory. (Types) `pwd`. Let's grab that so we get it exactly right. We'll grab a copy of that. Okay. Now I will head over to our UI where we will make some changes. Those changes are in the profile. Sensor profiles, as I said just one more time if we go to **Sensors** the most accurate way of doing it is to find our sensor and click here to edit it, we know we're guaranteed to get the right sensor. To get to the FIM profile for our Linux sensor server host where you go to **Agent**. We can see here FIM is off. We hit the slider to turn it on. What we can see here is that there are two pages: one and two down here, of default entries there: **/bin**, **/boot**, etcetera, and then on page two – similar.

02:00-03:00
The only other thing to point out here is–let's go through the check box is that this path is enabled. In this example, **/usr/sbin**. **Inclusions** is a star, so everything's included so everything in that folder, nothing's excluded. In the directory, it will go ten sub directories deep looking for any change, create, or deletions of files and directories in that subfolder. What we want to do is add what we just copied out. We'll go back to the beginning. We're

going to add a row, double-click in **Path**, and paste that in there. **Inclusions** we want everything in there. We don't want to exclude anything. Not that we'll need it for this example but the **Depth** is ten sub directories deep and we're going to do **Change**, **Create**, and **Delete**. After we've done that, click **Submit**.

03:00-04:00

We can see immediately that all the servers, the server sensors, that are using that profile–we've got two of them. AD Server three, the Windows one, the Linux one. a little refresh It shouldn't take long for that to come back. We are up and running again. Let's go back to our Linux host. Let's create a file in there. (Types) `touch david.txt`. Let's see it. There's our file. Let's also remove it. A couple of different actions happening in that directory that's been monitored. Okay. Let's see what we can see in the user interface. The first place we can look is **Visualize**. There's a dashboard for this: **Visualize** and **File Integrity Monitoring**. We can see we have a bunch in there. The ones we're looking for in particular, are our /home/scengineer/FileMonitor.

04:00-05:00

Make that a little larger. We can see there it is. There's our david.txt, and there were three events related to that. Alright. Let's see where else we can find information regarding this **Investigate | Threat Hunting**. We're going to change from **Alerts** to **Linux Events**. Here we are. We've got a bunch of Linux Events. Now the first thing you can see is under our (Top) **Log Source Classes** we have an FIM one in there. You could filter for those directly. We can still see down here we have a list of what's in the table. If I wanted to specifically look for what I was searching for, we could type a "search for." Let's type a search in for `file_path:`.

05:00-06:00

Now this could be a little tricky because we're going to use some RegEx expressions: backslash to say the next one is a literal, forward slash home, backslash literal forward slash, SC engineer, backslash, the next one is a forward slash is file monitor test, and finish with a quote. [`file_path: "\/home\/scengineer\/FileMonitorTest"`] Let's search for that. We have three. What we can see down here is the three. The path that they came from, so we know that was the path, the extension–what was the rule? The rule was **/home/scengineer/FileMonitorTest**. That came from our sensor profile. There should be one more that tells us about what was created, so let's open it up. The easiest way to find there. Click the "**i**." Go to the **Details** tab. We're going to look for a *file create* or *delete* in here. Here we have the **event_action** deleted.

06:00-07:00

Let's go look at a different one. Let's just do event action. If we can add that field as a column, we can open up here and look for event.action. There it is: event action delete. We're going to add that column, and then minimize this record. We can see all of them. We can see the created, attributes modified, and deleted. That's our file that we created, obviously a modification in the attributes of that file and then we remove that file. That's how we can use the **b** feature. Turn it on in the sensor profile, push it down, and of course you've got to get the path names correct. Then either look for this in the **Visualize** dashboard. We can go to our dashboard and look.

07:00-07:25
We should see only three now. Why are we only seeing three? There were so many more! It's because our search has been persistent across **Investigate | Threat Hunting** and our visualization in this dashboard.

The lesson, *Demonstration of Enhanced FIM,* is complete. Thank you.

L06 - ENHANCED NETWORK TRAFFIC

L06A-PRES: Enhanced Network Traffic (07m:25s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Enhanced Network Traffic. Enhanced Network Traffic.* The Linux Server sensor includes a network traffic monitoring feature similar to that of a Modular Sensor. This lesson will show the differences and also the configuration and validation of the network traffic feature.

This lesson is about network traffic. The Linux Server sensor, in a similar way to the Modular Sensor, is able to perform network traffic analysis. This lesson will look at the Linux Server sensor and the sensor profile to ingest the network traffic. Now when we say similar to a Modular Sensor, in this diagram we've got both a Linux Server and a Modular Sensor. If we look at the Modular Sensor for a moment we can see the Modular Sensor has two interfaces, a LAN interface, and a SPAN interface.

01:00-2:00
We can make the assumption that the SPAN interface is connected to the vSwitch LAN, and it's a port group that has promiscuous mode turned on. Any ethernet frames traversing that virtual switch LAN will be replicated to the port group SPAN and sent to the Modular Sensor. Now that will include traffic from the Windows server to the internet. We call that North-South traffic, or we could say any traffic from the Windows server to the Linux Server,

we call that East-West traffic. Any of that traffic would be included and replicated or mirrored if you like, to the SPAN port on the Modular Sensor. All that traffic would then be sent as Interflow records to the Data Processor.

02:00-03:00

The difference with the Linux Server for a network traffic perspective and why we say it's similar is because it does collect traffic, but it will only collect North-South traffic from the Linux Server itself. Only the server that the server sends a software is loaded on when that Linux Server communicates to either the Windows server, East-West traffic, or to the internet (North-South traffic) because it's the Linux Server that's communicating. It's really North-South traffic for it. It's leaving the Linux Server and hitting the switch. We're really looking at only that traffic, as opposed to the Modular Sensor, which will see all of the traffic. There's no specific requirement on our switch or our port group in this example where an ESXi configuration, no need for that special configuration. This is the Linux Server sensor software that is monitoring the network traffic entering and leaving that Linux host particularly.

03:00-04:00

How do we compare it? If we go back to our Linux Server sensor, we can see here we have run `aella_cli` and then we have run a `show interface` command. Now I don't know if you remember when we go back to the very beginning when we first installed this software, and we did a `show interface` to confirm an IP address of that management–that 10.20.1.23 address, that was the only interface because up until then it didn't even have a profile that was pushed down. The moment the profile is pushed down that's been bound to this sensor, we'll see that we've now got an ethernet zero. It's ethernet zero that's the interesting interface to look at from a network traffic perspective. What I want you to look at particularly with ethernet zero is at the bottom there, RX bytes I want you to pay attention.

04:00-05:00

We call this the North-South traffic or the traffic coming to and from that Linux host itself. We count that traffic there. If we were to go into the user interface to look at this, we could go to **System | Sensors**. Let's look where the network traffic is configured. Much the same way we would configure this for a Modular Sensor with its profile, we're going to configure this for the Linux Server sensor also in its profile. But let's look first. We can see there that it's using the profile Training Server Sensor, so we click on that to edit. Okay. In this tab, we can see that there's a **Network** tab. That **Network** tab is what is related to the Linux network traffic, that sensor profile for network traffic. You can see there by default, it's got Application Identification, Application Session, Application Metadata, Process Correlation, Handshake Failure, and Flood Attack.

05:00-06:00
Those are all on by default. That's why we actually didn't have to do anything to see that traffic appearing in our ethernet zeros. It's already tracking, when we look there. If you are to make changes, so if you were to expand one of those and look at some of the more granular configuration changes we could make in there, you can expand the feature, you could make those changes. But if you do make those changes, make sure to select **Submit** to push those changes down to our Linux Server sensor. Okay. After we've done that, let's go have a look. We're going to navigate to **System** and **Sensors** and we're going to look at our sensor table. We're going to find our Linux Server sensor name. Notice in those columns there is now **Input**, well, it's not "now," it was always there, **Input** and **Output** columns. They are the ones that are worth counting to see the traffic coming in and out of that Linux Server.

06:00-07:00
You'll see those increase as you see the traffic coming to and from that server. The other place to look for it, is if you click the details of that sensor, that server sensor, the Linux Server sensor, we can see here we're looking at our ublnx-sj-tr-dbw3. We can see there, there is also an **Input** and **Output** there. We can see the ingestion there as well. The other place is in **Investigate | Threat Hunting**. Change the index from **Alert** to **Traffic**. Then we can see there we're going to do a filter for our type of sensor, which is a Linux agent in this example. When we do that we can see already that we have one of our devices, our Linux Server sensor, by the name ublnx-sj-tr-dbw3.

07:00-07:25
We can see we're in the **Traffic** index. This is traffic being reported from that sensor. They are Interflow records of the metadata of the traffic coming into the Data Processor.

The lesson, *Enhanced Network Traffic*, is complete. Thank you.

L06B-DEMO: Enhanced Network Traffic (07m:00s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Linux Server Sensor Installation and Configuration* module is *Demonstration of Enhanced Network Traffic. Demonstration of Enhanced Network Traffic.* The Linux Server sensor includes a network traffic monitoring feature similar to that of a Modular Sensor. This lesson will demonstrate the differences and also the configuration and validation of the Network Traffic feature.

In this demonstration, I will show you the Network Traffic feature for the Linux Server sensor. We'll look at where it's configured and how it works and do some testing to validate the

configuration. Let's start with the Linux Server sensor command line. Here we are in our Linux Server. Let's open up the sensor. Type `aella_cli`. The first thing we want to do is a `show interface`.

01:00-02:00

We could do `show int` and tab. There we go. We were looking at the `show interface` command to look at the management IP address of that 10.20.1.23, which is the interface address of this server. When we do a `show interface` command, because we've added a sensor profile to this Linux sensor, we can now see we've got additional interface. The one I want to pay particular attention to is ethernet zero and particularly RX bytes. We can see there it's currently 32.1 megabytes. I want you to try and remember that as we go through and look at the features and we'll generate some traffic to try and increase that. Let's go look in the UI. In the UI let's look where we can do that. **System | Sensors**. I find it easy to do this. Look at our host, our profile.

02:00-03:00

The first tab, the **Network** tab, for this profile that our Linux Server sensor is bound to has the Network Traffic features: Application Identification, Application Session, Application Metadata. If we were to expand one of these we can see that there are more granular configuration items that we can change in here. This is where we would change its set up. These are all defaults. I haven't changed anything, but if you were to change anything it would be a **Submit** to continue. We can just close that. Now go look at where we can monitor some of this. If we open the detail of our sensor, we can see down here we've got **Interflow Stats**. This 33.9M input and the 765.6k output. The difference between that is of course we're collecting the metadata in our Interflow record, not the actual data. 33 meg has been monitored, and the metadata of that is 765.

03:00-04:00

I think if you do the calculations, that's where we end up with the Interflow reduction ratio of 44 times to one is what's actually been ingested into our Data Lake. We can also go go into **Investigate | Threat Hunting**, change from **Alerts**. We're not going to look at Linux Events. We're going to look at **Traffic** because that's the index that this would go to. Let's do a filter. We can see for our log source. Let's filter for the Linux agent or we can actually come up here and filter for our actual host for that sensor. Filter for with the plus, and now we're looking just at our server. We see the **Traffic** index over here. Remember it was about thirty three point something. Let's run a speed test on our host to increase some of that traffic.

04:00-05:00

Back to our host. Let's quit out of that CLI. We're back here and we're going to run a speed test CLI. This will just generate some traffic. It's going to download a file, push a file back up,

test the speed. What we can remember here this was 32. That's something we want to remember while this test runs. The download's done. Now the upload's happening. Just wait for those to finish and then we'll go back into our sensor CLI and a `show interface`. We can just see there we've gone from 32.1 to 185.2. Let's jump into our UI and look for those same settings. The first place we'll go, let's clear our filters, **Clear all**.

05:00-06:00

We're going to go to **System**, **Sensors**, and look at our sensor. There we go. Now we're at 182.7. That's jumped right up. Clearly what they downloaded is very compressible, so a lot of data and of course we're only taking the metadata information into our Data Lake. **Investigate | Threat Hunting**, change from **Alerts** to **Traffic**. This time, instead of choosing our sensor name, we'll just go with **Linux Agent**. Filter for, and we can see that that's jumped up right there. Alright. That's how we can clarify that we're looking at...it's only the North-South traffic. If we just go back to that PowerPoint that we were looking at earlier, we know that the Modular Sensor is looking at SPAN traffic and that's all the traffic in and out of this switch, East-West traffic.

06:00-07:00

If these Windows hosts were talking to the Linux hosts, that's East-West traffic, and it'd be mirrored to the SPAN port into our Modular Sensor. The North-South traffic would be the Windows or Linux servers communicating through the switch, through the firewall, and out through to the internet. That's our North-South traffic. Now we'd say well why are we not seeing the traffic from this Windows server? Well, all we're really seeing is that link there. The Linux Server to the switch. It's like it's got a network traffic monitor or a SPAN port just for its interface. That's where we're looking there. It's not a SPAN traffic monitor, like we have in our Modular Sensor. It is just for the North-South in and out of that Linux Server itself.

The lesson, *Demonstration of Enhanced Network Traffic*, is complete. Thank you.